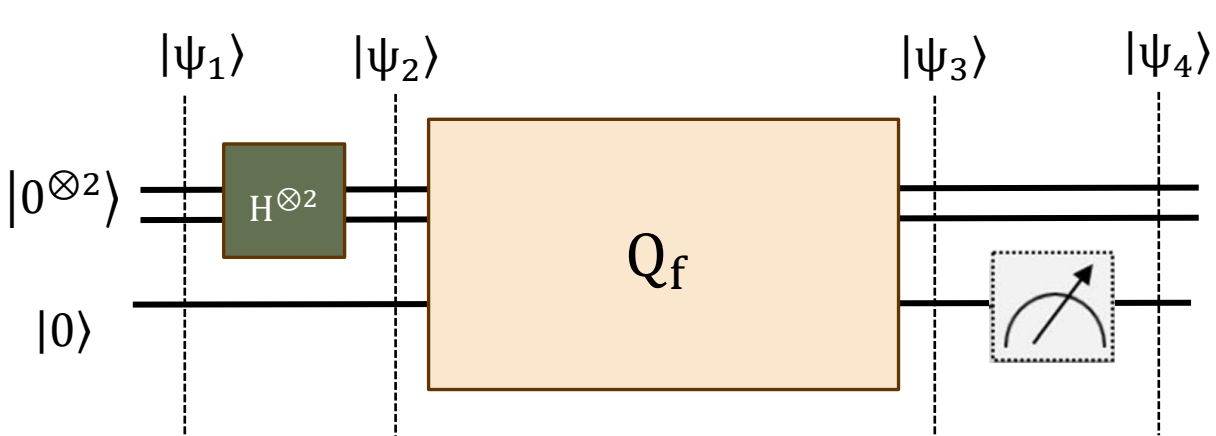


Introduction to Quantum Computing (L-7)

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Rules for Partial Measurement of Quantum Registers



$$|\psi_3\rangle = \frac{|00\rangle|0\rangle + |01\rangle|1\rangle + |10\rangle|0\rangle + |11\rangle|1\rangle}{2}$$

Notice: First register values are correlated with second register

Suppose after measuring the 2nd register, we get $|0\rangle$ at the output.

Then only the term $\frac{|00\rangle + |10\rangle}{\sqrt{2}}$ will be the surviving content of the 1st register at $|\psi_4\rangle$

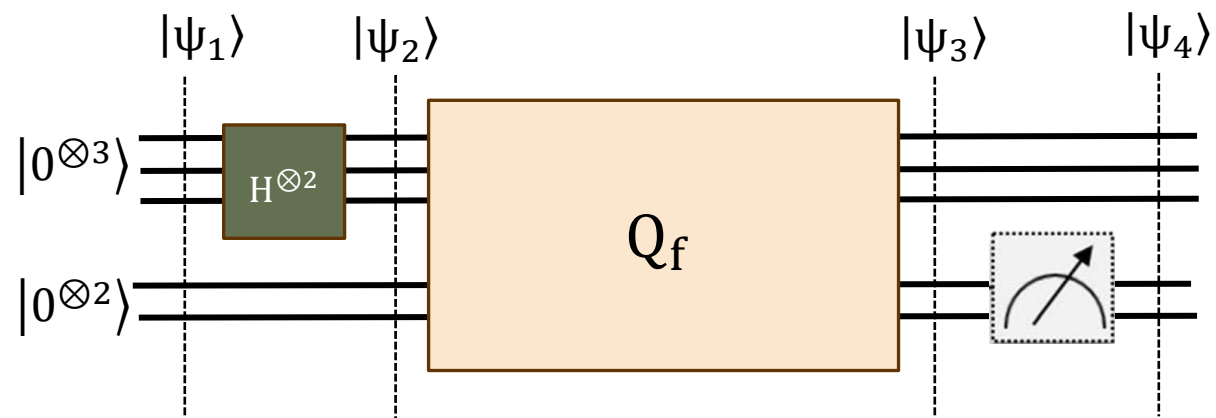
Suppose after measuring the 2nd register, we get $|1\rangle$ at the output.

Then only the term $\frac{|10\rangle + |11\rangle}{\sqrt{2}}$ will be the surviving content of the 1st register at $|\psi_4\rangle$

Partial measurement:

- Does NOT destroy the entire state
- It only collapses the measured register (the 2nd register in the above example)
- The unmeasured register (the 1st register in the above example) becomes a structured partial superposition

Rules for Partial Measurement of Quantum Registers



$$|\psi_3\rangle = \frac{|000\rangle|00\rangle + |011\rangle|01\rangle + |100\rangle|00\rangle + |111\rangle|01\rangle}{2}$$

Suppose after measuring the 2nd register, we get 00 at the output.

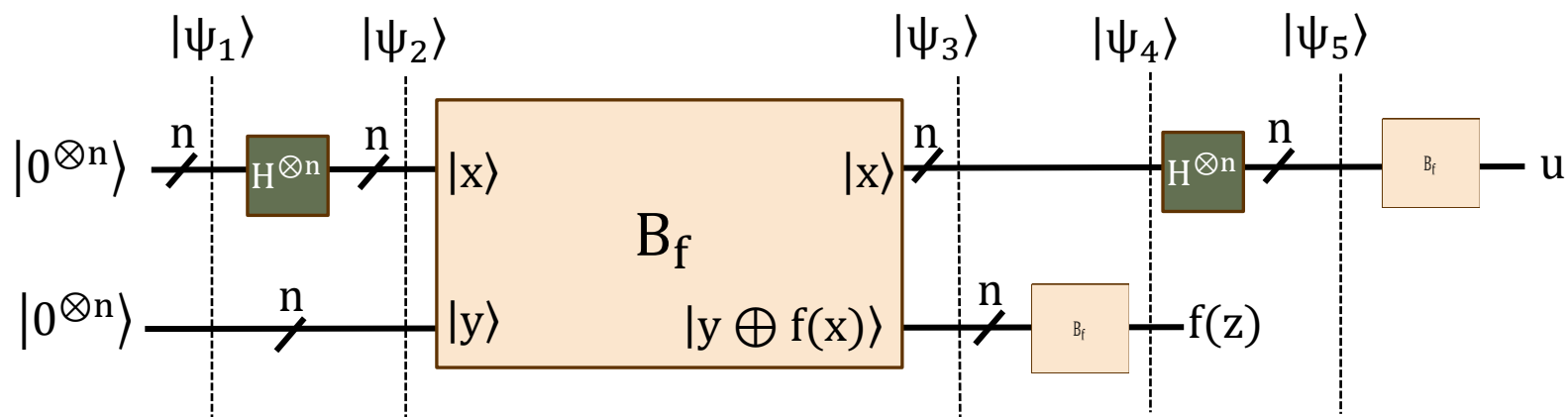
The state of the first register at $|\psi_4\rangle$ will be $\frac{|000\rangle + |100\rangle}{\sqrt{2}}$

Suppose we run the circuit the 2nd time, and this time we measure the after measuring the 2nd register, we get 01 at the output.

The state of the first register at $|\psi_4\rangle$ will be $\frac{|011\rangle + |111\rangle}{\sqrt{2}}$

Note: At $|\psi_4\rangle$ we are no longer writing the state of the second register, since it has already been measured. $|\psi_4\rangle$ is simply whatever is the state of the first register

Recap Simon's Algorithm



- Problem: Given a function $f(x)$ with the promise $f(x) = f(x \oplus s)$, find the hidden non-zero string s .
- Quantum Advantage: Finds s using $O(n)$ queries, whereas classical algorithms require exponential queries.
- Key Idea: Quantum superposition + interference gives measurement outcomes u satisfying $u \cdot s = 0 \pmod{2}$
- Classical Post-Processing: Solve linear equations mod 2 to determine s .
- Why It Matters (Application): Simon's algorithm introduced the idea of hidden periodicity detection, which later led to Shor's factoring algorithm

Quantum Fourier Transform

Primitive roots of unity: $Z^n = 1; Z \in \mathbb{C}$

The equation will have n roots (solutions): $\omega_n^0, \omega_n^1, \omega_n^2, \dots, \omega_n^{n-1}$

$\omega_n^k = e^{2\pi i k/n}$ **ω_n^k lie on a unit circle in a complex plane**

$$|\omega_n^k| = \sqrt{(\omega_n^k)^* \omega_n^k} = \sqrt{e^{-2\pi i k/n} \cdot e^{2\pi i k/n}} = \sqrt{e^0} = 1$$

ω_n^k is a periodic function, that is $\omega_n^k = \omega_n^{k \bmod n}$

Example: Find the primitive roots of unity: $Z^4 = 1$

The primitive roots are: $\omega_4^0, \omega_4^1, \omega_4^2, \omega_4^3$

$$= e^{2\pi i 0/4}, e^{2\pi i 1/4}, e^{2\pi i 2/4}, e^{2\pi i 3/4}$$

So, the primitive roots of unity: $Z^4 = 1$ are 1, i , -1 , and $-i$

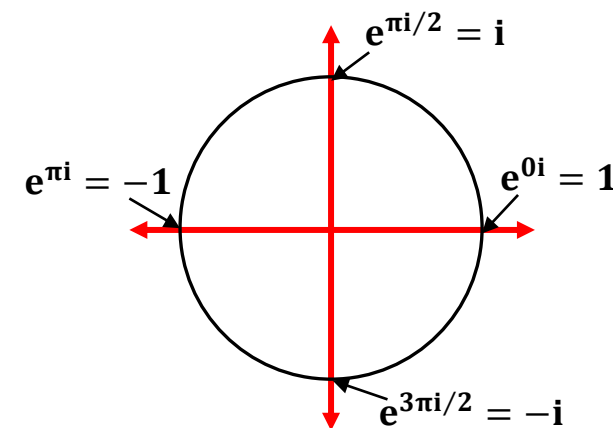
Remember ω_n^k is a periodic function, that is $\omega_n^k = \omega_n^{k \bmod n}$

$$\omega_4^4 = \omega_n^{4 \bmod 4} = \omega_4^0 = 1$$

$$\omega_4^5 = \omega_n^{5 \bmod 4} = \omega_4^1 = i$$

$$\omega_4^6 = \omega_n^{6 \bmod 4} = \omega_4^2 = -1$$

$$\omega_4^7 = \omega_n^{7 \bmod 4} = \omega_4^3 = -i$$



Remember: $e^{i\theta} = \cos\theta + i\sin\theta$

Remember: $e^{i\theta}$ lie on a unit circle on an complex plane at an angle θ

Quantum Fourier Transform

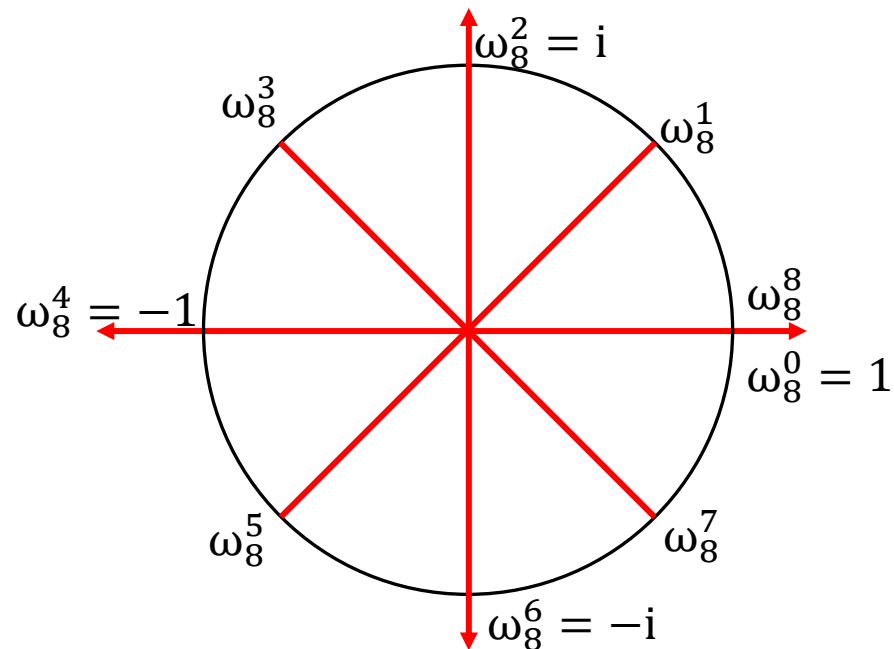
Example: Find the primitive roots of unity: $Z^8 = 1$

The primitive roots are: $\omega_8^0, \omega_8^1, \omega_8^2, \omega_8^3, \dots, \omega_8^7$

$$= e^{2\pi i 0/8}, e^{2\pi i 1/8}, e^{2\pi i 2/8}, e^{2\pi i 3/8} \dots e^{2\pi i 7/8}$$

Instead of this, we can directly go to our unit circle on a complex plane and look at the roots pictorially

Now we are finding the 8th root of unity, the unit circle will be divided into 8 parts of $360/8 = 45^\circ$ quadrant each



Find the primitive roots of unity: Z^8 are

$$1, \omega_8^1, i, \omega_8^3, -1, \omega_8^5, -i, \omega_8^7$$

$$\omega_8^3 = \omega_8^1 \cdot \omega_8^2 = i\omega_8^1$$

$$\omega_8^5 = -\omega_8^1$$

$$\omega_8^7 = -\omega_8^3 = -i\omega_8^1$$

$$\omega_n^k = e^{2\pi i k/n}$$

$$\begin{aligned}\omega_8^1 &= e^{2\pi i 1/8} = e^{i\pi/4} = \sin(\pi/4) + i\cos(\pi/4) \\ &= \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}(1 + i)\end{aligned}$$

$$\omega_8^3 = i\frac{1}{\sqrt{2}}(1 + i) = \frac{1}{\sqrt{2}}(i - 1)$$

$$\omega_8^5 = -\omega_8^1 = -\frac{1}{\sqrt{2}}(1 + i)$$

$$\omega_8^7 = -\omega_8^3 = \frac{1}{\sqrt{2}}(1 - i)$$

Quantum Fourier Transform

QFT Matrix is $F_n = \frac{1}{\sqrt{n}}$

$$\begin{pmatrix} \omega_n^{0 \cdot 0} & \omega_n^{0 \cdot 1} & \omega_n^{0 \cdot 2} & \dots & \omega_n^{0 \cdot (n-1)} \\ \omega_n^{1 \cdot 0} & \omega_n^{1 \cdot 1} & \omega_n^{1 \cdot 2} & \dots & \omega_n^{1 \cdot (n-1)} \\ \omega_n^{2 \cdot 0} & \omega_n^{2 \cdot 1} & \omega_n^{2 \cdot 2} & \dots & \omega_n^{2 \cdot (n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \omega_n^{(n-1) \cdot 0} & \omega_n^{(n-1) \cdot 1} & \omega_n^{(n-1) \cdot 2} & \dots & \omega_n^{(n-1) \cdot (n-1)} \end{pmatrix}$$

$\begin{matrix} \text{0}^{\text{th}} \text{ col} & \text{1}^{\text{st}} \text{ col} & \text{2}^{\text{nd}} \text{ col} & & \text{(n-1)}^{\text{th}} \text{ col} \end{matrix}$

$\begin{matrix} \text{0}^{\text{th}} \text{ row} \\ \text{1}^{\text{st}} \text{ row} \\ \text{2}^{\text{nd}} \text{ row} \\ & \vdots \\ \text{(n-1)}^{\text{th}} \text{ row} \end{matrix}$

In a QFT matrix, remember, we start counting the rows and columns from 0, instead of 1.

It's a matrix, whose entry of the u^{th} row and v^{th} column is $\omega_n^{u \cdot v}$

$$(F_n)_{u,v} = \frac{1}{\sqrt{n}} \omega_n^{u \cdot v}, \quad u, v = 0, 1, \dots, (n-1)$$

Normalization $1/\sqrt{n}$ makes F_n unitary: $F_n^\dagger F_n = I$

Write the QFT matrix F_4

$$F_4 = \frac{1}{2} \begin{bmatrix} \omega_4^{0 \cdot 0} & \omega_4^{0 \cdot 1} & \omega_4^{0 \cdot 2} & \omega_4^{0 \cdot 3} \\ \omega_4^{1 \cdot 0} & \omega_4^{1 \cdot 1} & \omega_4^{1 \cdot 2} & \omega_4^{1 \cdot 3} \\ \omega_4^{2 \cdot 0} & \omega_4^{2 \cdot 1} & \omega_4^{2 \cdot 2} & \omega_4^{2 \cdot 3} \\ \omega_4^{3 \cdot 0} & \omega_4^{3 \cdot 1} & \omega_4^{3 \cdot 2} & \omega_4^{3 \cdot 3} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \omega_4^0 & \omega_4^0 & \omega_4^0 & \omega_4^0 \\ \omega_4^0 & \omega_4^1 & \omega_4^2 & \omega_4^3 \\ \omega_4^0 & \omega_4^2 & \omega_4^4 & \omega_4^6 \\ \omega_4^0 & \omega_4^3 & \omega_4^6 & \omega_4^9 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega_4^1 & \omega_4^2 & \omega_4^3 \\ 1 & \omega_4^2 & \omega_4^0 & \omega_4^2 \\ 1 & \omega_4^3 & \omega_4^2 & \omega_4^1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$$

Quantum Fourier Transform

We have seen $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

When we move to two qubits, the state space becomes 4-dimensional. The basis states are tensor products:

$$\begin{aligned} |00\rangle &= |0\rangle \otimes |0\rangle; |01\rangle = |0\rangle \otimes |1\rangle; \\ |10\rangle &= |1\rangle \otimes |0\rangle; |11\rangle = |1\rangle \otimes |1\rangle \end{aligned}$$

$$\begin{aligned} |00\rangle &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} & |01\rangle &= \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} & |10\rangle &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} & |11\rangle &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad |\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$F_4|\psi\rangle = |\varphi\rangle$. Find $|\varphi\rangle$

$$F_4 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$$

$$F_4|\psi\rangle = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2\sqrt{2}} \left\{ 1 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix} + 0 \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ -i \\ -1 \\ i \end{bmatrix} \right\}$$

$$|\varphi\rangle = \frac{1}{2\sqrt{2}} \begin{bmatrix} 2 \\ 1-i \\ 0 \\ 1+i \end{bmatrix} = \frac{1}{2\sqrt{2}} \left\{ 2 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + (1-i) \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + (1+i) \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

After QFT, if we measure $|\varphi\rangle$

We shall measure '00' with a probability $|1/\sqrt{2}|^2 = 1/2$

We shall measure '01' with a probability $|(1-i)/2\sqrt{2}|^2$

We shall measure '10' with a probability $|0|^2$

We shall measure '11' with a probability $|(1+i)/2\sqrt{2}|^2$

Quantum Fourier Transform

$$F_4 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$$

$$c_0 = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad c_1 = \frac{1}{2} \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix}$$

$$c_2 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \quad c_3 = \frac{1}{2} \begin{bmatrix} 1 \\ -i \\ -1 \\ i \end{bmatrix}$$

$$\|c_0\|^2 = \frac{1 + 1 + 1 + 1}{4} = 1 \quad \|c_3\|^2 = \frac{1 + |-i|^2 + 1 + |i|^2}{4} = 1$$

Any column c_v of a QFT matrix is normalized

$$c_0^\dagger c_1 = \frac{1 \cdot 1 + 1 \cdot i + 1 \cdot (-1) + 1 \cdot (-i)}{4} = 0$$

$$c_2^\dagger c_3 = \frac{1 \cdot 1 + (-1) \cdot (-i) + 1 \cdot (-1) + (-1) \cdot i}{4} = 0$$

Distinct columns of the QFT matrix are orthogonal.

The columns of the F_4 form an orthonormal set.

In fact, columns of any F_n will always form an orthonormal set

$$c_u^\dagger c_v = \begin{cases} 1 & \text{when } u = v \\ 0 & \text{when } u \neq v \end{cases}$$

This ensures that $F_n^\dagger F_n = I$. A matrix is unitary if and only if its columns form an orthonormal basis.

This is why unitarity matters and every quantum gate must be unitary:

- Orthonormal columns $\rightarrow$ lengths preserved
- Length preservation $\rightarrow$ total probability conserved
- Probability conservation $\rightarrow$ physically allowed quantum evolution
- Unitary matrices are exactly the transformations that do not destroy or create probability.

Quantum Fourier Transform

$$F_n = \frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega_n^1 & \omega_n^2 & \dots & \omega_n^{n-1} \\ 1 & \omega_n^2 & \omega_n^4 & \dots & \omega_n^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega_n^{n-1} & \omega_n^{2(n-1)} & \dots & \omega_n^{(n-1)(n-1)} \end{bmatrix}$$

Let F_n acts on an input vector say $\begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{n-1} \end{bmatrix}$ and produces an output $\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{bmatrix}$

$$\frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega_n^1 & \omega_n^2 & \dots & \omega_n^{n-1} \\ 1 & \omega_n^2 & \omega_n^4 & \dots & \omega_n^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega_n^{n-1} & \omega_n^{2(n-1)} & \dots & \omega_n^{(n-1)(n-1)} \end{bmatrix} \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{n-1} \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{bmatrix}$$

The probability of measuring an output state $|k\rangle$ is simply $|\beta_k|^2$

Now if there is any linear shift in the input vector by 1, the output vector stays only the same, but gathers a phase

$$\frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega_n^1 & \omega_n^2 & \dots & \omega_n^{n-1} \\ 1 & \omega_n^2 & \omega_n^4 & \dots & \omega_n^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega_n^{n-1} & \omega_n^{2(n-1)} & \dots & \omega_n^{(n-1)(n-1)} \end{bmatrix} \begin{bmatrix} \alpha_{n-1} \\ \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{n-2} \end{bmatrix} = \begin{bmatrix} \omega_n^0 \beta_0 \\ \omega_n^1 \beta_1 \\ \omega_n^2 \beta_2 \\ \vdots \\ \omega_n^{n-1} \beta_{n-1} \end{bmatrix}$$

The probability of measuring an output state $|k\rangle$ is simply $|\omega_n^k \beta_k|^2 = |\omega_n^k|^2 |\beta_k|^2$

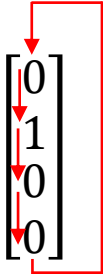
$$|\omega_n^k| = \sqrt{(\omega_n^k)^* \omega_n^k} = \sqrt{e^{-2\pi i k/n} \cdot e^{2\pi i k/n}} = \sqrt{e^0} = 1$$

So, the probability is still $|\beta_k|^2$

QFT turns shifts in time (input index) into phases in frequency and quantum measurement ignores phase. This single property will be explicitly used in period finding algorithms.

Quantum Fourier Transform

$$F_4 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} \quad |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$



$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |10\rangle$$

$$F_4|01\rangle = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix}$$

$$F_4|10\rangle = \frac{1}{2} \begin{bmatrix} \omega_4^0 1 \\ \omega_4^1 i \\ \omega_4^2 (-1) \\ \omega_4^3 (-i) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \cdot 1 \\ i \cdot i \\ -1 \cdot (-1) \\ -i \cdot (-i) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

- We measure '00' with a probability $|1/2|^2 = 1/4$
- We measure '01' with a probability $|1/2|^2 |i|^2 = 1/4$
- We measure '10' with a probability $|-1/2|^2 = 1/4$
- We measure '11' with a probability $|1/2|^2 |-i|^2 = 1/4$

- We measure '00' with a probability $|1/2|^2 = 1/4$
- We measure '01' with a probability $|-1/2|^2 = 1/4$
- We measure '10' with a probability $|1/2|^2 = 1/4$
- We measure '11' with a probability $|-1/2|^2 = 1/4$

Shift in input index → Phase factors in output

But:

Quantum measurement ignores phase → probabilities stay the same.

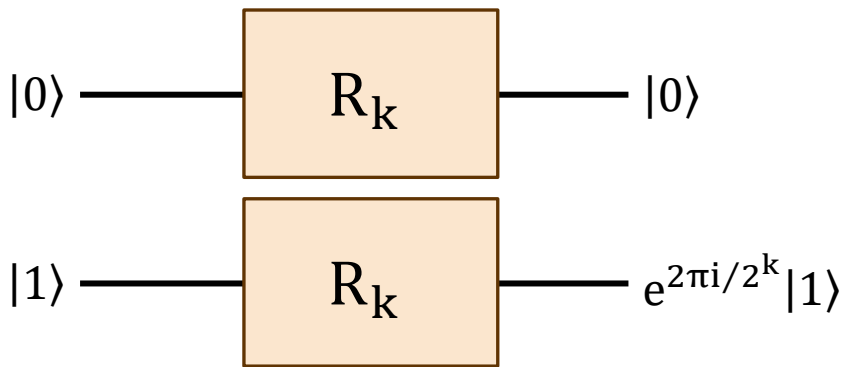
Quantum Fourier Transform

$$F_n \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{r-1} \\ \alpha_0 \\ \alpha_1 \\ \vdots \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{(N/r)-1} \\ \beta_0 \\ \beta_1 \\ \vdots \end{bmatrix}$$

Let QFT on a function $f(x)$ lead to an output of $g(x)$: $\text{QFT } f(x) = g(x)$

If $f(x)$ has a periodicity of ' r ' then $g(x)$ will have a periodicity of N/r where $N = 2^n$

The R_k gate: The R_k gate induces a rotation of a single-qubit about the z-axis:



$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i / 2^k} \end{bmatrix}$$

It induces a phase rotation by $2\pi/2^k$ radians of the $|1\rangle$ component of an arbitrary qubit while leaving the $|0\rangle$ component unchanged

$$R_1 = \begin{bmatrix} 1 & 0 \\ 0 & e^{\pi i} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

So, $R_1|1\rangle = -|1\rangle$. Its just a 'Z' gate

$$R_2 = \begin{bmatrix} 1 & 0 \\ 0 & e^{\pi i / 2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

$R_2|1\rangle = i|1\rangle$. Acquired a phase of $+\pi/2$ radians

$$R_3 = \begin{bmatrix} 1 & 0 \\ 0 & e^{\pi i / 4} \end{bmatrix}$$

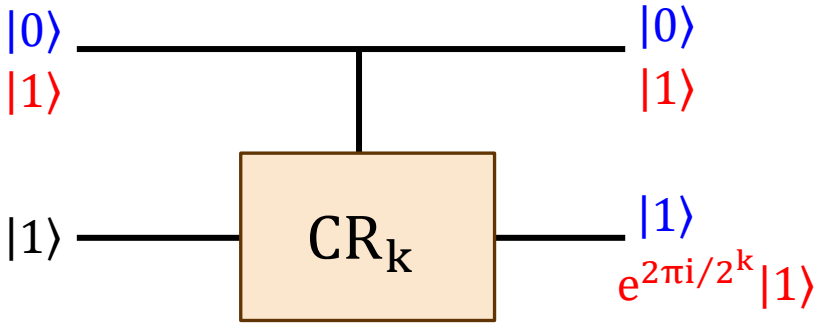
$R_2|1\rangle = e^{\pi i / 4}|1\rangle$. Acquired a phase of $+\pi/4$ radians

R_k does change the probability of measuring $|1\rangle$. It only rotates its phase. Also, remember that phase changes are invisible to measurement unless they interfere with something else.

Quantum Fourier Transform

The Controlled R_k gate: A controlled R_k gate applies the R_k gate if and only if the control qubit is $|1\rangle$

If the control qubit is $|1\rangle$, the CR_k gate rotates the phase of the target qubit's $|1\rangle$ component by $2\pi/2^k$ radians . Otherwise, it does nothing.



$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$CR_k|\psi\rangle = \frac{|00\rangle + e^{2\pi i/2^k}|11\rangle}{\sqrt{2}}$$

$$CR_2|\psi\rangle = \frac{|00\rangle + i|11\rangle}{\sqrt{2}}$$

$$|11\rangle \xrightarrow{CR_k} e^{2\pi i/2^k} |11\rangle R_3|\psi\rangle = \frac{|00\rangle + e^{\pi i/4}|11\rangle}{\sqrt{2}}$$

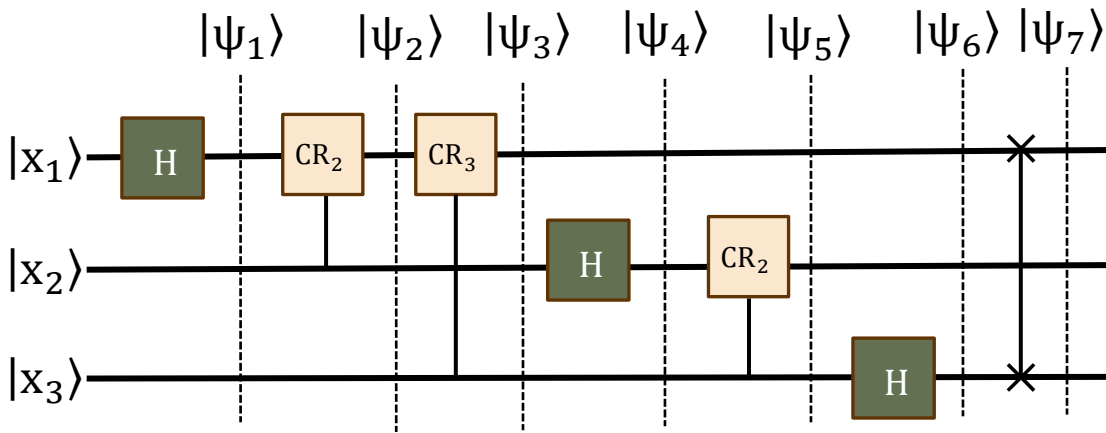
In the computational basis of $|00\rangle, |01\rangle, |10\rangle,$ and $|11\rangle$

$$CR_k = \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{2\pi i/2^k} \end{bmatrix} \begin{matrix} |00\rangle \xrightarrow{CR_k} |00\rangle \\ |01\rangle \xrightarrow{CR_k} |01\rangle \\ |10\rangle \xrightarrow{CR_k} |10\rangle \\ |11\rangle \xrightarrow{CR_k} e^{2\pi i/2^k} |11\rangle \end{matrix}$$

Only $|11\rangle$ will pick up the phase

- CR_k lets one qubit write its value into the phase of another qubit (QFT stores information in phases, not amplitudes).
- Each CR_k adds the correct fractional phase $2\pi/2^k$ needed for the Fourier transform. (Binary digits become binary fractions).
- Without CR_k , QFT would only create a superposition, not frequency information (Hadamard alone is not enough).

Quantum Fourier Transform



Lets take $|x_1x_2x_3\rangle = |101\rangle$

Remember that in the computational basis of $\{|0\rangle, |1\rangle\}$, $|101\rangle =$

Evaluate $F_8|101\rangle$

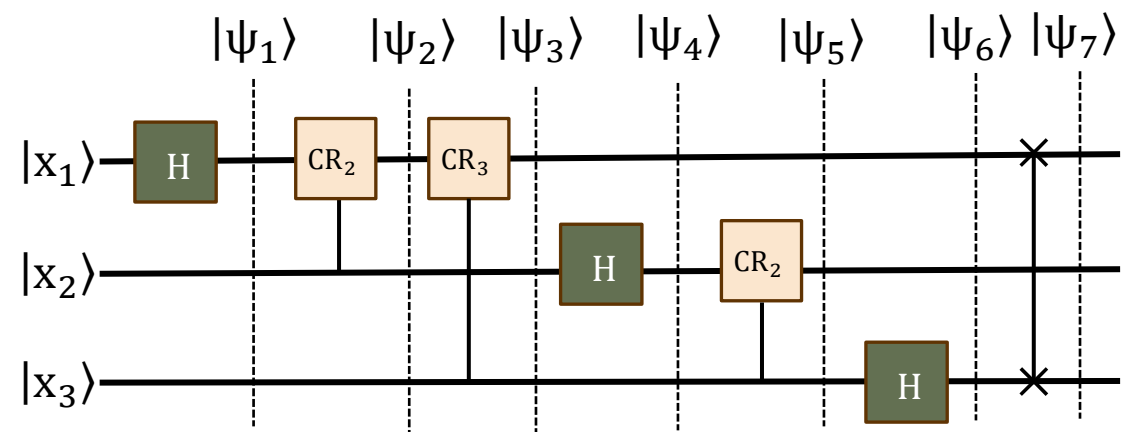
$$F_8 = \frac{1}{\sqrt{8}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \omega^4 & \omega^5 & \omega^6 & \omega^7 \\ 1 & \omega^2 & \omega^4 & \omega^6 & 1 & \omega^2 & \omega^4 & \omega^6 \\ 1 & \omega^3 & \omega^6 & \omega & \omega^4 & \omega^7 & \omega^2 & \omega^5 \\ 1 & \omega^4 & 1 & \omega^4 & 1 & \omega^4 & 1 & \omega^4 \\ 1 & \omega^5 & \omega^2 & \omega^7 & \omega^4 & \omega & \omega^6 & \omega^3 \\ 1 & \omega^6 & \omega^4 & \omega^2 & 1 & \omega^6 & \omega^4 & \omega^2 \\ 1 & \omega^7 & \omega^6 & \omega^5 & \omega^4 & \omega^3 & \omega^2 & \omega \end{pmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$F_8|101\rangle = \frac{1}{\sqrt{8}} \begin{bmatrix} 1 \\ \omega^5 \\ \omega^2 \\ \omega^7 \\ \omega^4 \\ \omega^1 \\ \omega^6 \\ \omega^3 \end{bmatrix} = \frac{1}{\sqrt{8}} \begin{bmatrix} 1 \\ e^{2\pi i .5/8} \\ e^{2\pi i .2/8} \\ e^{2\pi i .7/8} \\ e^{2\pi i .4/8} \\ e^{2\pi i .1/8} \\ e^{2\pi i .6/8} \\ e^{2\pi i .3/8} \end{bmatrix}$$

Recall: $\omega_n^k = e^{2\pi i k/n}$

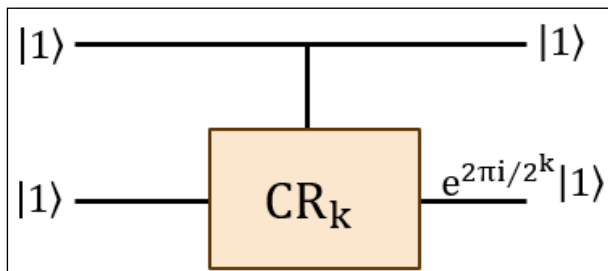
Quantum Fourier Transform



$$H|x\rangle = \frac{|0\rangle + e^{2\pi i x/2} |1\rangle}{\sqrt{2}}$$

$$H|0\rangle = \frac{|0\rangle + e^0 |1\rangle}{\sqrt{2}} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|1\rangle = \frac{|0\rangle + e^{\pi i/2} |1\rangle}{\sqrt{2}} = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$



$$|\psi_1\rangle = H|x_1\rangle|x_2x_3\rangle = H|1\rangle|01\rangle = \frac{|0\rangle + e^{\pi i} |1\rangle}{\sqrt{2}} |01\rangle$$

$$|\psi_2\rangle = CR_2|\psi_1\rangle = |\psi_1\rangle \text{ (Since the control qubit is '0')}$$

$$|\psi_3\rangle = CR_3|\psi_2\rangle = \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\frac{2\pi i}{8}} e^{\pi i} |1\rangle \right] |01\rangle$$

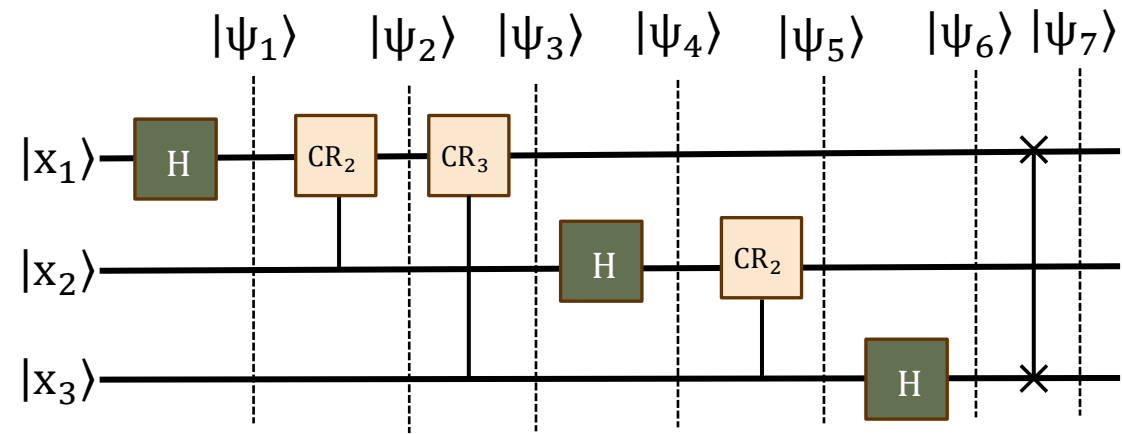
$$= \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\frac{2\pi i 5}{8}} |1\rangle \right] |01\rangle = |q_1\rangle |01\rangle$$

$$|\psi_4\rangle = |q_1\rangle H|01\rangle = |q_1\rangle \frac{|0\rangle + e^0 |1\rangle}{\sqrt{2}} |1\rangle$$

$$|\psi_5\rangle = |q_1\rangle CR_2 \frac{|0\rangle + |1\rangle}{\sqrt{2}} |1\rangle = |q_1\rangle \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\frac{\pi i}{2}} |1\rangle \right] |1\rangle = |q_1\rangle |q_2\rangle |1\rangle$$

$$|\psi_6\rangle = |q_1\rangle |q_2\rangle H|1\rangle = |q_1\rangle |q_2\rangle \frac{|0\rangle + e^{\pi i/2} |1\rangle}{\sqrt{2}}$$

Quantum Fourier Transform



$$|\psi_6\rangle = |q_1\rangle|q_2\rangle H|1\rangle = |q_1\rangle|q_2\rangle \frac{|0\rangle + e^{\pi i/2}|1\rangle}{\sqrt{2}}$$

$$|q_1\rangle = \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\frac{2\pi i 5}{8}}|1\rangle \right]$$

$$|q_2\rangle = \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\frac{\pi i}{2}}|1\rangle \right]$$

$$|\psi_6\rangle = \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\frac{2\pi i 5}{8}}|1\rangle \right] \quad \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\frac{\pi i}{2}}|1\rangle \right] \quad \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\pi i/2}|1\rangle \right]$$

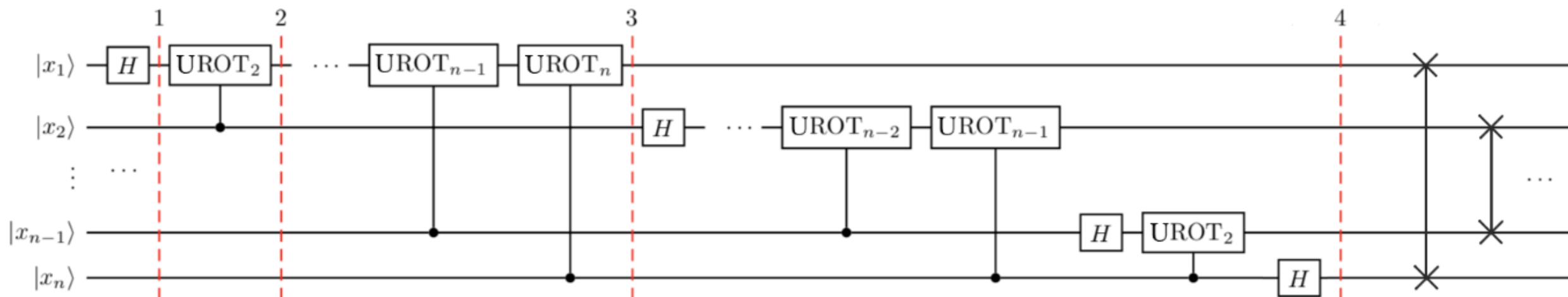
$$|\psi_7\rangle = \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\pi i/2}|1\rangle \right] \quad \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\frac{\pi i}{2}}|1\rangle \right] \quad \frac{1}{\sqrt{2}} \left[|0\rangle + e^{\frac{2\pi i 5}{8}}|1\rangle \right]$$

$$|\psi_7\rangle = \frac{1}{\sqrt{8}} \left[|000\rangle + e^{\frac{2\pi i .5}{8}}|001\rangle + e^{\frac{2\pi i}{4}}|010\rangle + \dots e^{\frac{2\pi i .3}{8}}|111\rangle \right]$$

$$|\psi_7\rangle = \frac{1}{\sqrt{8}} \begin{bmatrix} 1 \\ e^{2\pi i .5/8} \\ e^{2\pi i .2/8} \\ e^{2\pi i .7/8} \\ e^{2\pi i .4/8} \\ e^{2\pi i .1/8} \\ e^{2\pi i .6/8} \\ e^{2\pi i .3/8} \end{bmatrix}$$

This confirms (through example) that the circuit implements F_8 .
You are encouraged to verify this yourself:
Try other inputs such as $|011\rangle$, $|010\rangle$, or $|111\rangle$.

Quantum Fourier Transform

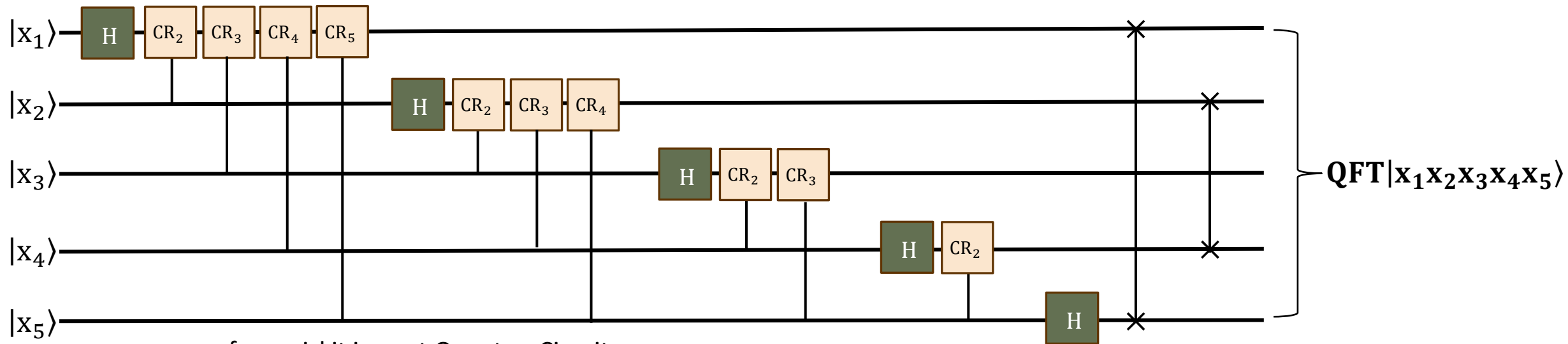


- The circuit uses only:
Hadamard gates
Controlled phase rotations (UROT / controlled- R_k)
SWAP gates

$$F_{2^n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i x k / 2^n} |k\rangle$$

- QFT does not compute Fourier coefficients numerically. It encodes them into quantum phases.
- Phases are accumulated from more significant qubits to less significant ones. Each qubit “writes” its binary value into the phase of lower qubits.

Quantum Fourier Transform



```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector
import numpy as np
n = 5
qc = QuantumCircuit(n)
for target in range(n):
    qc.h(target)
    for control in range(target + 1, n):
        angle = np.pi / (2 * (control - target))
        qc.cp(angle, control, target)
for i in range(n // 2):
    qc.swap(i, n - i - 1)
qc.draw('mpl')
state = Statevector.from_instruction(qc)
```

Period Finding Algorithm

Given a periodic function $f: \{0,1\}^n \rightarrow \{0,1\}^n$ find the period r such that $f(x) = f(x + kr)$ where k is any integer except 0

For demonstration let us choose $n = 4$ and $f(x) = x \pmod 3$

The algorithm will check that $f(0) = 0; f(1) = 1; f(2) = 2; f(3) = 0$

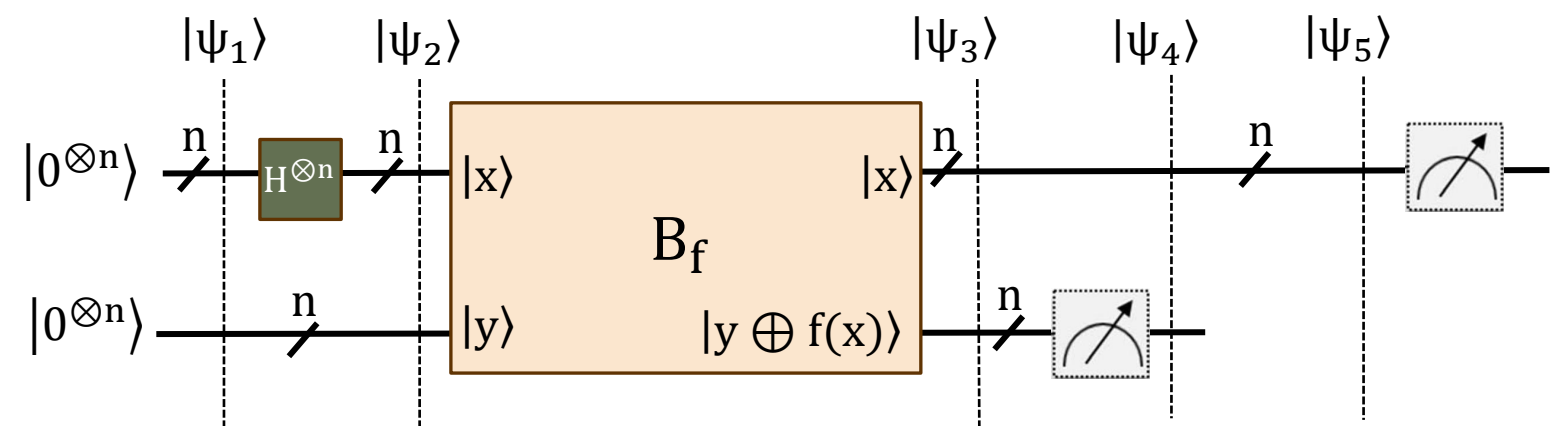
The moment it finds the same output, it will evaluate the period r as $|3 - 0| = 3$

Classical it finds the period in $O(r)$ time complexity. But note, we choose this example where $r = 3$. In general, r can be as large as the number of inputs, which is 2^n for an n -bit input string.

In general the time complexity of a classical period finding algorithm is $O(2^n)$, which is exponential.

$ x\rangle_2$	$ x\rangle_{10}$	$f(x)$
0000	0	0
0001	1	1
0010	2	2
0011	3	0
0100	4	1
0101	5	2
0110	6	0
0111	7	⋮
1000	8	
1001	9	
1010	10	
1011	11	
1100	12	
1101	13	
1110	14	
1111	15	

Period Finding Algorithm



Oracle: $B_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$
 $B_f|x\rangle|0\rangle = |x\rangle|f(x)\rangle$

We want the input to the oracle to be not just any specific input ‘x’ but rather all possible inputs to make the use of quantum superposition. So, use Hadamard

$$H^{\otimes n} |0^{\otimes n}\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=\{0,1\}^n} |x\rangle$$

Since we have all possible inputs ‘x’, the output of the oracle at the 2nd register will have all possible values of f(x)

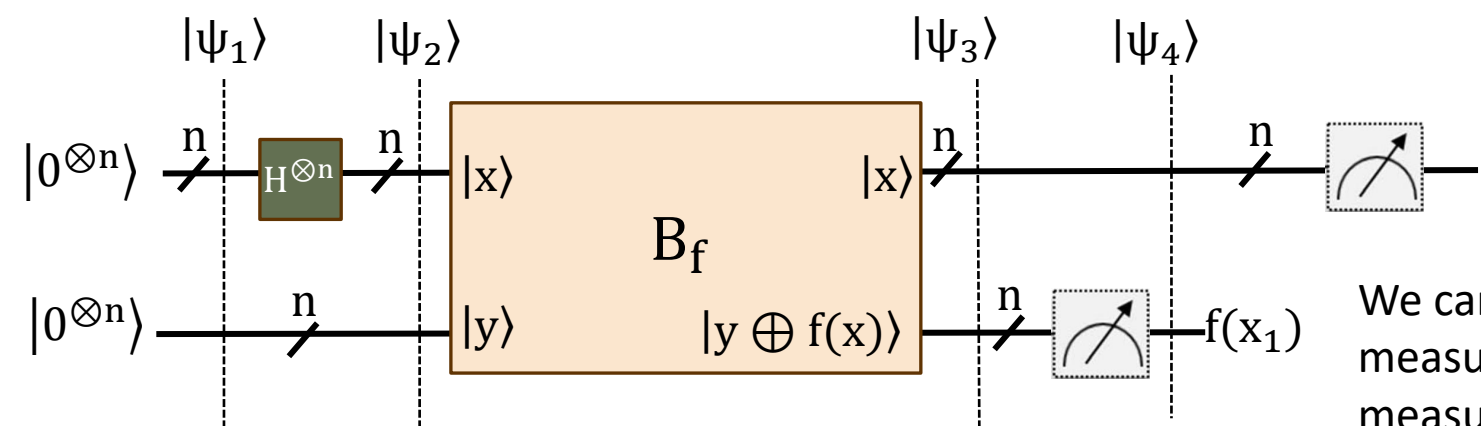
$$\frac{1}{\sqrt{2^n}} \sum_{x=\{0,1\}^n} |f(x)\rangle$$

|ψ₃⟩ is a tensor product between the content of the 1st register and the content of the 2nd register.

$$\frac{1}{\sqrt{2^n}} \sum_{x=\{0,1\}^n} |x\rangle \frac{1}{\sqrt{2^n}} \sum_{x=\{0,1\}^n} |f(x)\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=\{0,1\}^n} |x\rangle |f(x)\rangle$$

Lets us now measure the 2nd register and get some specific value say f(x₁). Then at |ψ₄⟩, the 1st register will contain only those values of x, that can leads to f(x₁) at the 2nd register. Also note, we have not measured the 1st register yet, so all these values of x will be at equal superposition at the 1st register

Period Finding Algorithm



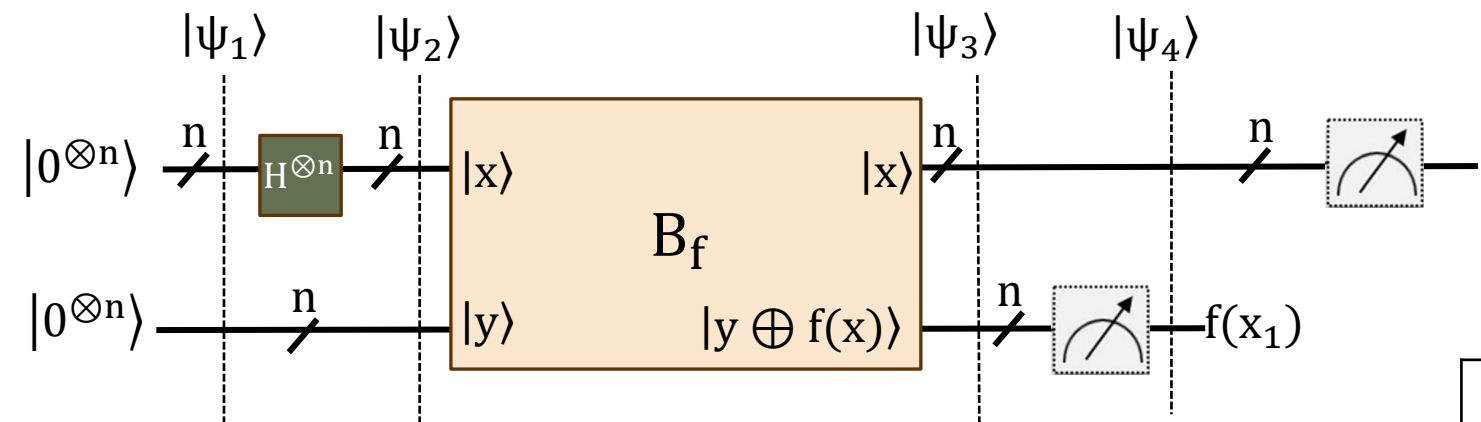
Since the function is periodic, at $|\psi_4\rangle$ the 1st register shall not have $|x_1\rangle$ but also $|x_1 + r\rangle, |x_1 + 2r\rangle, |x_1 + 3r\rangle\dots$ and they will all be in equal superposition

$$\frac{|x_1\rangle + |x_1 + r\rangle + |x_1 + 2r\rangle + \dots + |x_1 + \{(2^n/r) - 1\}r\rangle}{\sqrt{2^n/r}}$$

We can try to measure the 1st register. The moment we measure a register we loose the superposition and the measurement collapses to a specific pure state, say x_1 . We re-run the circuit, and hope that the next time we get to measure $x_1 + 2r$, run it again and measure $x_1 + 5r$ and therefore by running the circuit a few times, we gather enough data to evaluate the periodicity ‘r’.

During the 2nd run of the circuit, the measurement of the 2nd register may not be $f(x_1)$. There is not guarantee that it will always be $f(x_2)$ or $f(x_7)$ or some other value. The $|\psi_4\rangle$ at the 1st register will therefore renormalize at each run based on the content of the 2nd register. So we cannot get any information on r

Period Finding Algorithm

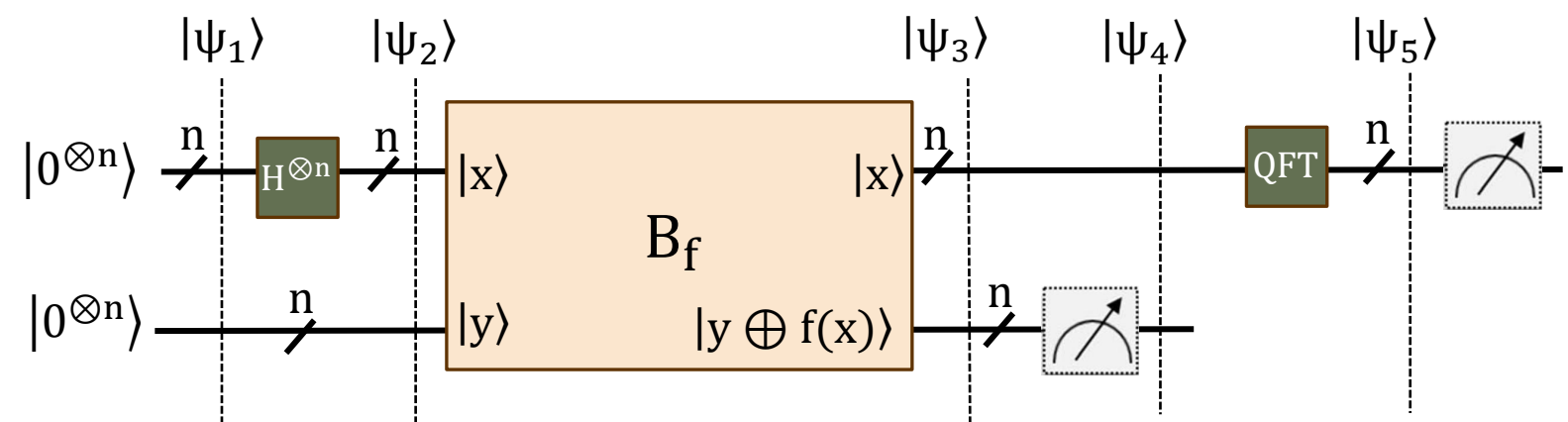


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2 nd register Measured	1 st register superposition at $ \psi_4\rangle$
$f(0)$	$\frac{ 0\rangle + 0+r\rangle + 0+2r\rangle + \dots}{\sqrt{2^n/r}}$
$f(1)$	$\frac{ 1\rangle + 1+r\rangle + 1+2r\rangle + \dots}{\sqrt{2^n/r}}$
$f(2)$	$\frac{ 2\rangle + 2+r\rangle + 2+2r\rangle + \dots}{\sqrt{2^n/r}}$
$\vdots$	$\vdots$
$f(k)$	$\frac{ k\rangle + k+r\rangle + k+2r\rangle + \dots}{\sqrt{2^n/r}}$

Period Finding Algorithm



Why Do We Insert a QFT After|ψ₄>?

After measuring the 2nd register, the 1st register is a periodic superposition: $\frac{|k\rangle + |k + r\rangle + |k + 2r\rangle + \dots}{\sqrt{2^n/r}}$

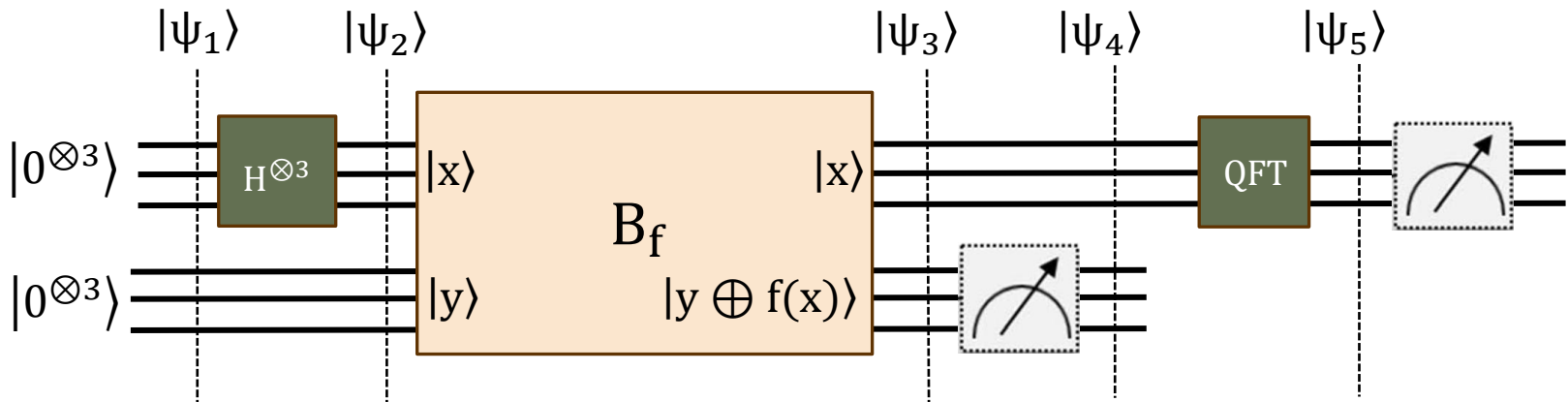
k could be 0,1,2....etc. The input to the QFT is **periodic with period r**. But, the period is hidden in the spacing of the states.

QFT turns regular spacing in position into regular spacing in frequency.
In simple words, a shift in the input becomes a phase in the output. Shift by r $\xrightarrow{\text{QFT}_n}$ $e^{2\pi ikr/2^n}$

Period r in input $\xrightarrow{\text{QFT}_n}$ Period $\frac{2^n}{r}$ after QFT

|ψ₅> will expect amplitudes concentrated at $k = 0, \left(\frac{2^n}{r}\right), 2\left(\frac{2^n}{r}\right), 3\left(\frac{2^n}{r}\right), \dots$

Period Finding Algorithm



Let $f: \{0,1\}^3 \rightarrow \{0,1\}^3$ find the period r such that $f(x) = x(\text{mod}2)$

$$|\psi_1\rangle = |000\rangle|000\rangle \quad |\psi_2\rangle = H^{\otimes 3}|000\rangle|000\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=\{0,1\}^3} |x\rangle|000\rangle$$

$$|\psi_2\rangle = \frac{|0\rangle + |1\rangle + |2\rangle + |3\rangle + |4\rangle + |5\rangle + |6\rangle + |7\rangle}{\sqrt{8}} |000\rangle$$

$$\begin{aligned} |\psi_3\rangle &= B_f|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=\{0,1\}^3} |x\rangle|f(x)\rangle \\ &= \frac{|0\rangle|f(0)\rangle + |1\rangle|f(1)\rangle + |2\rangle|f(2)\rangle \dots + |7\rangle|f(7)\rangle}{\sqrt{8}} \\ &= \frac{|0\rangle|0\rangle + |1\rangle|1\rangle + |2\rangle|0\rangle + |3\rangle|1\rangle + |4\rangle|0\rangle + |5\rangle|1\rangle + |6\rangle|0\rangle + |7\rangle|1\rangle}{\sqrt{8}} \end{aligned}$$

x	f(x) = x (mod2)
000 = 0	0
001 = 1	1
010 = 2	0
011 = 3	1
100 = 4	0
101 = 5	1
110 = 6	0
111 = 7	1

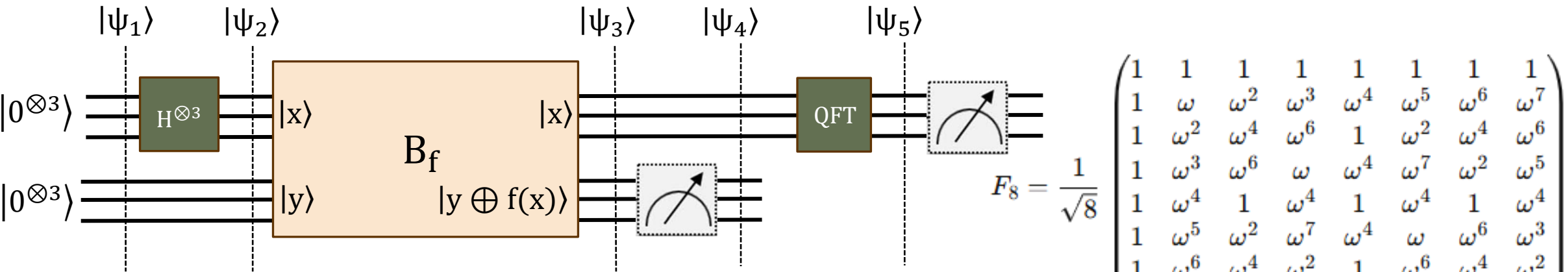
Lets say that as we measure the 2nd register we get 0. Then by rule of partial measurement, the state at the 1st register must be

$$\frac{|0\rangle + |2\rangle + |4\rangle + |6\rangle}{\sqrt{4}}$$

If instead we measure the 2nd register we get 1. Then the state at the 1st register must be

$$\frac{|1\rangle + |3\rangle + |5\rangle + |7\rangle}{\sqrt{4}}$$

Period Finding Algorithm



$$|\psi_5\rangle = \text{QFT}_8|\psi_4\rangle$$

Assume that we measure $|1\rangle$ at the 2nd. So the input to the QFT block is

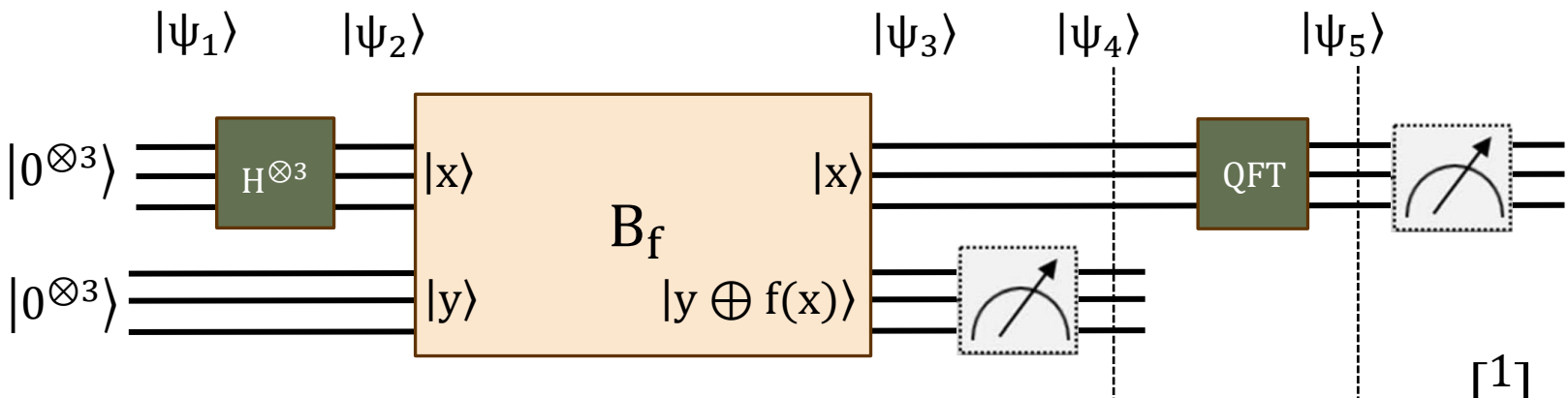
$$\frac{|1\rangle + |3\rangle + |5\rangle + |7\rangle}{\sqrt{4}} = \frac{1}{\sqrt{4}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{8}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & e^{2\pi i/8} & i & e^{6\pi i/8} & -1 & e^{10\pi i/8} & -i & e^{14\pi i/8} \\ 1 & i & -1 & -i & 1 & i & -1 & -i \\ 1 & e^{6\pi i/8} & -i & e^{2\pi i/8} & -1 & e^{14\pi i/8} & i & e^{10\pi i/8} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & e^{10\pi i/8} & i & e^{14\pi i/8} & -1 & e^{2\pi i/8} & -i & e^{6\pi i/8} \\ 1 & -i & -1 & i & 1 & -i & -1 & i \\ 1 & e^{14\pi i/8} & -i & e^{10\pi i/8} & -1 & e^{6\pi i/8} & i & e^{2\pi i/8} \end{pmatrix}$$

$$|\psi_5\rangle = \text{QFT}_8 \frac{1}{\sqrt{4}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{4\sqrt{2}} \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \frac{|0\rangle - |4\rangle}{\sqrt{2}}$$

After measurement, we get $|0\rangle$ with probability of $|1/\sqrt{2}|^2 = 1/2$ and $|4\rangle$ with probability of $|-1/\sqrt{2}|^2 = 1/2$

Period Finding Algorithm



If we had instead measured $|0\rangle$ at the 2nd register. So the input to the QFT block would be $\frac{|0\rangle + |2\rangle + |4\rangle + |6\rangle}{\sqrt{4}} = \frac{1}{\sqrt{4}}$

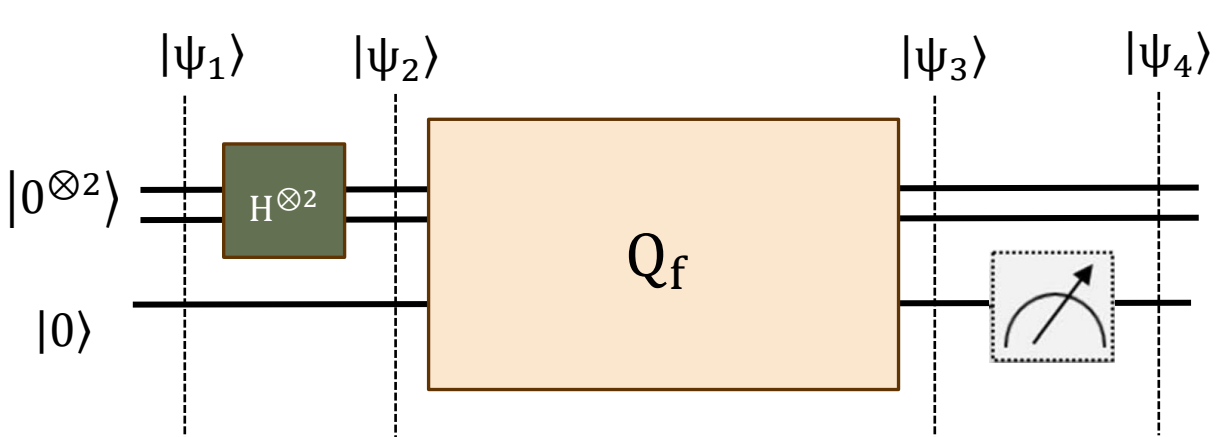
$$|\psi_5\rangle = \text{QFT}_8 \frac{1}{\sqrt{4}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \frac{|0\rangle + |4\rangle}{\sqrt{2}}$$

After measurement, we would still get $|0\rangle$ with probability of $|1/\sqrt{2}|^2 = 1/2$ and $|4\rangle$ with probability of $|1/\sqrt{2}|^2 = 1/2$

We have to re-run the circuit $\log 2^n (= \log 8$ times in this specific case) and each time we need to save the measured output of the 1st register. Then the GCD (greatest common divisor) of all these measured results in the periodicity of the measured values, which is $2^n/r$. From this the original periodicity of the input (r) is calculated. From saving onwards to the calculation of r is all classical (nothing quantum therein!)

In this example GCD of 0 and 4 is 4. This is the period of the output which is $2^n/r = 2^3/r$. So $r = 2^3/4 = 2$

Summary



$$|\psi_3\rangle = \frac{|00\rangle|0\rangle + |01\rangle|1\rangle + |10\rangle|0\rangle + |11\rangle|1\rangle}{2}$$

Notice: First register values are correlated with second register

Suppose after measuring the 2nd register, we get $|0\rangle$ at the output.

Then only the term $\frac{|00\rangle + |10\rangle}{\sqrt{2}}$ will be the surviving content of the 1st register at $|\psi_4\rangle$

Suppose after measuring the 2nd register, we get $|1\rangle$ at the output.

Then only the term $\frac{|10\rangle + |11\rangle}{\sqrt{2}}$ will be the surviving content of the 1st register at $|\psi_4\rangle$

Partial measurement:

- Does NOT destroy the entire state
- It only collapses the measured register (the 2nd register in the above example)
- The unmeasured register (the 1st register in the above example) becomes a structured partial superposition